

CSCE 378: Human Computer Interaction

Lecture: HCI & Usability Goals

AUGUST 27TH – ALISHA BEVINS

Today

1. What is HCI and UCD
2. P0 Document
3. Usability Goals
4. User Experience Goals
5. Introduction Survey Time

Announcements

- P0 due Friday by 11:59pm
- Introduction survey due end of class – complete before leaving
- If still need a group (or more members in group), come to front of room after class

Think of the last time you were frustrated using technology.

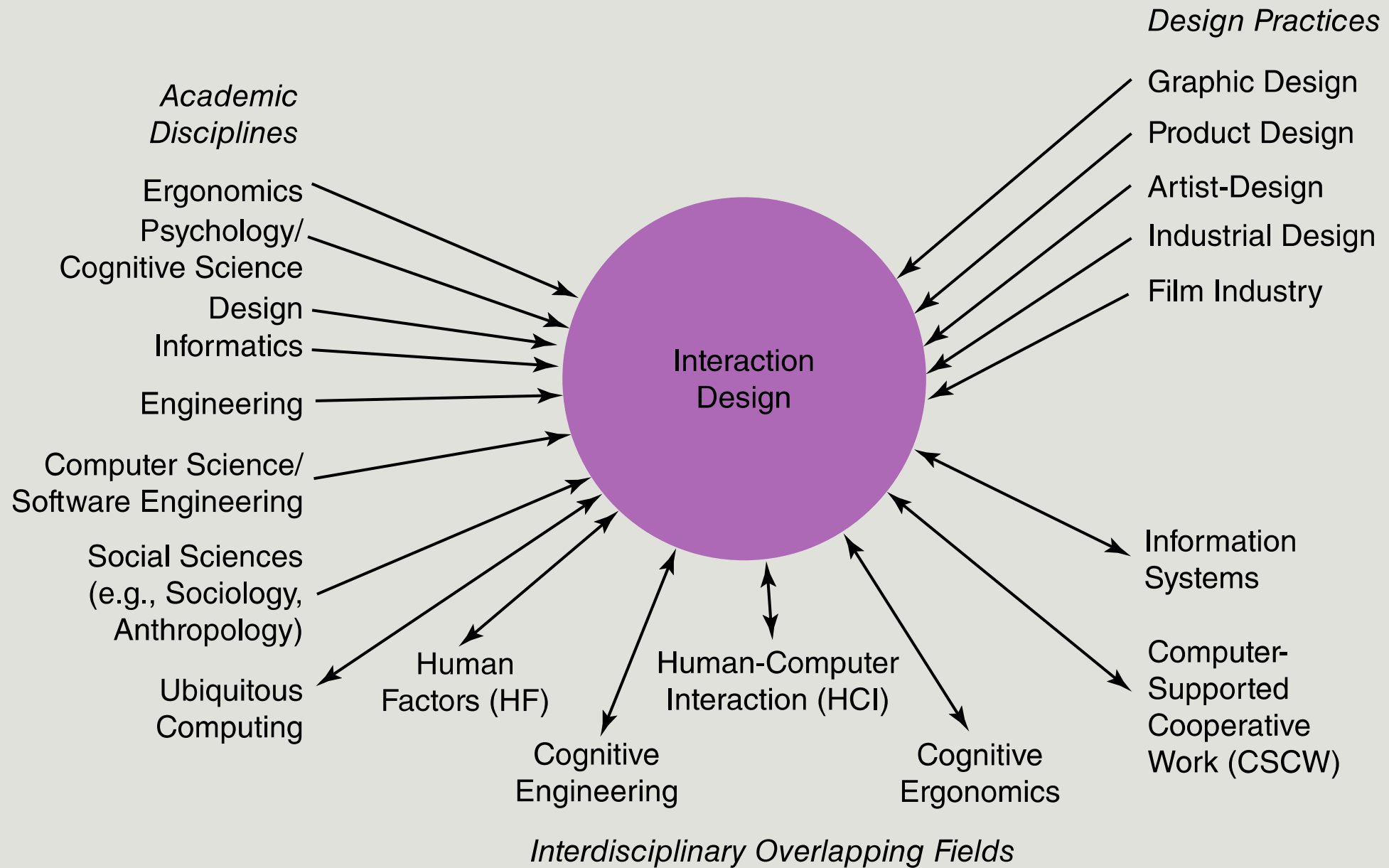
What happened?

What is HCI?

Human-Computer Interaction: Study of how people interact with technology

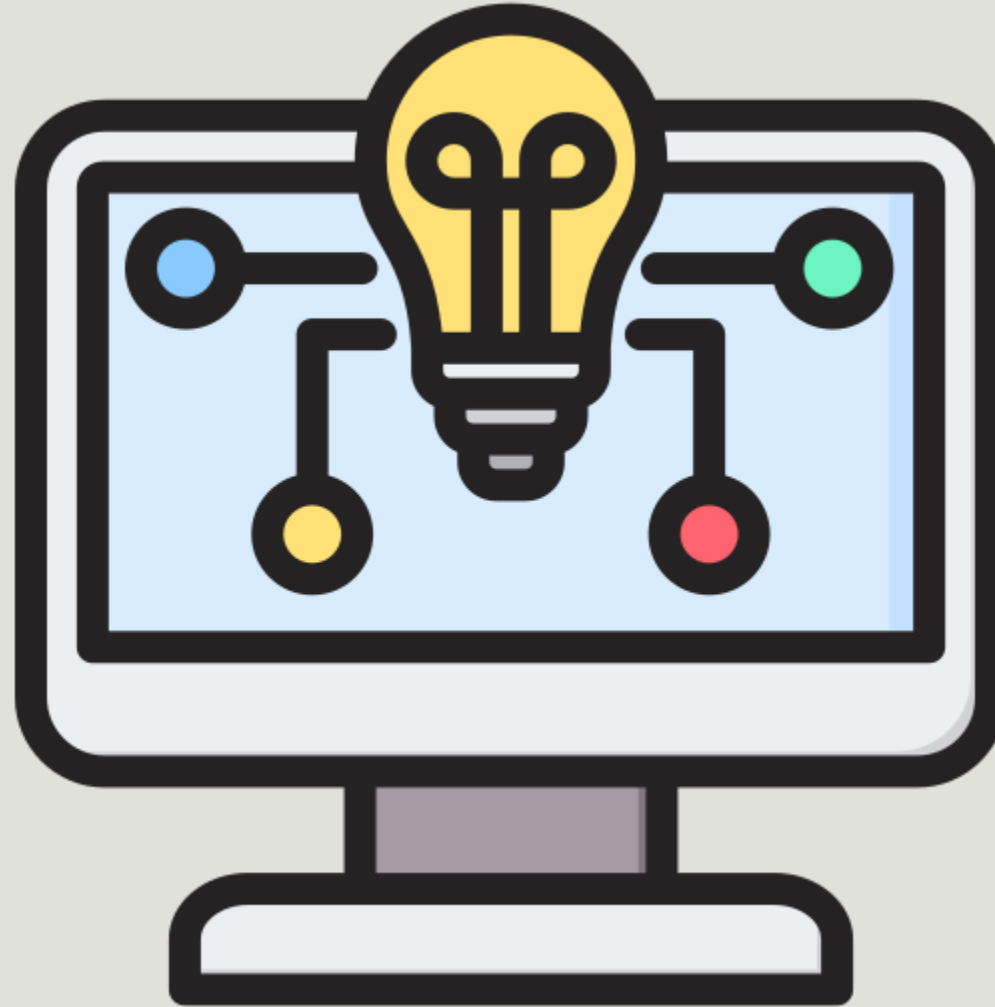
Three parts: User – Computer – Interface

Goal: Systems that are useful, usable, enjoyable



Computer Science/
Software Engineering

Design



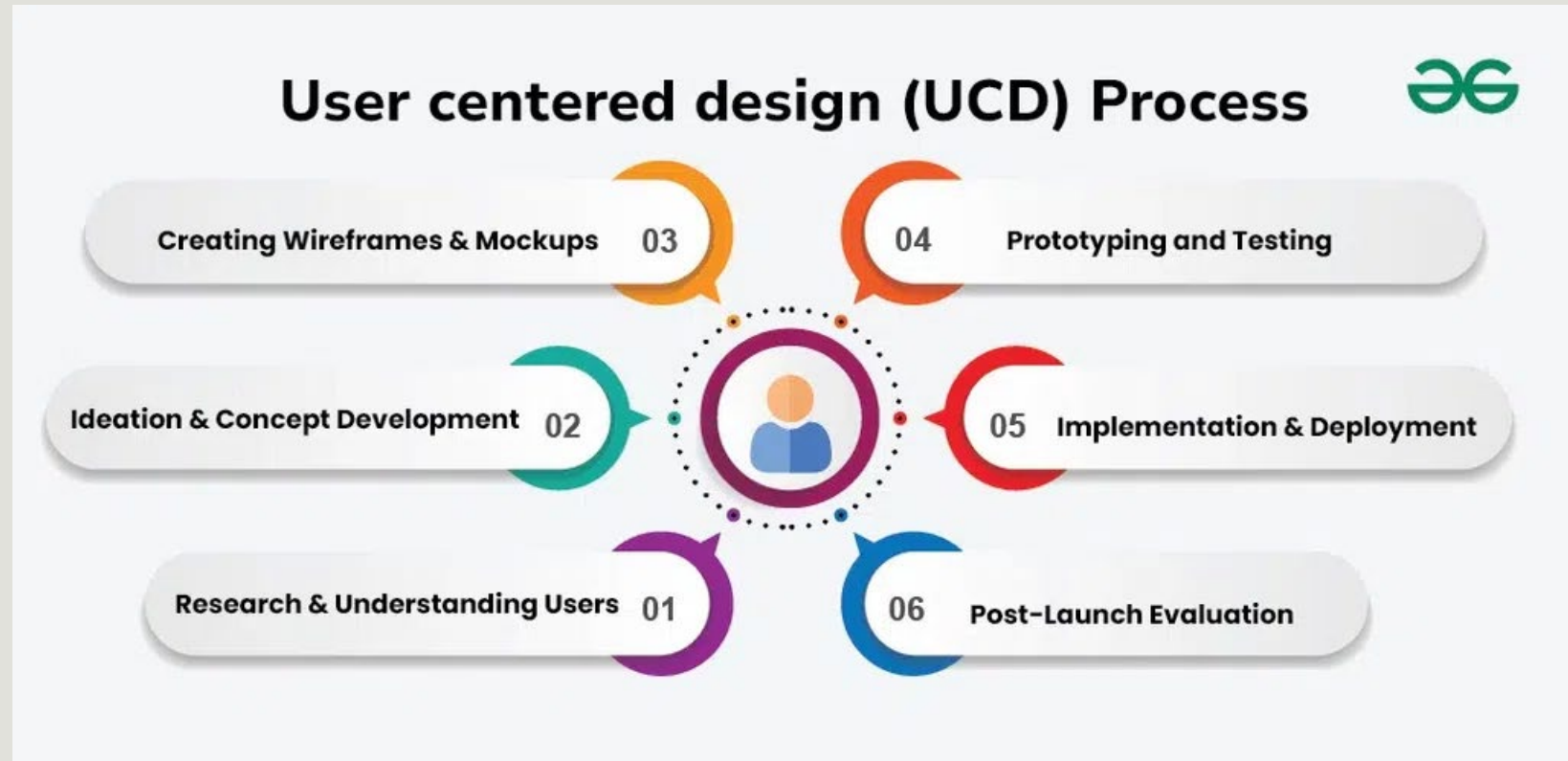
Psychology

Sociology



What is UCD?

- Focus on users at every stage
- Iterative & evidence-based



Why use UCD?

UCD

Involves users early

Direct contact with users

Continual user feedback

Designers **ARE NOT** users!

- One size does not fit all
- Helps to become aware of incorrect assumptions they may have about particular user groups (ex: not all elderly want or need big fonts)

User Goals

What are the user's goals?

Accomplish a task

How does the computer interface facilitate the user attaining the goals?

Transparency

P0 Document

P0 Framework

Project Team (Team Name)

Team Member Information

- Name of Member 1
 - Major
 - Bio (paragraph)
- Name of Member 2
 - Major
 - Bio (paragraph)
- ...

Mutual Areas of Interest (1 or 2 paragraphs)



“Good” vs “Bad” Design

- Form following function
- Emergence of ergonomics
- UX driven design practices



Why does it matter?

Commodity or Company Front?

- Remote = primary interface between the user and the brand
- Companies needed to deliver a positive experience
 - Focused on delivering the most positive experience possible
 - Anticipating users' needs before they do
 - Offering a solution that simplifies complexity



Consistency



3 User Types

Novice → Casual → Expert



Usability Goals

Usability = how **easy & pleasant** features are to use

Nielsen:

1. Learnability
2. Efficiency
3. Memorability
4. Errors (safety)
5. Satisfaction
 - Effectiveness
 - Utility

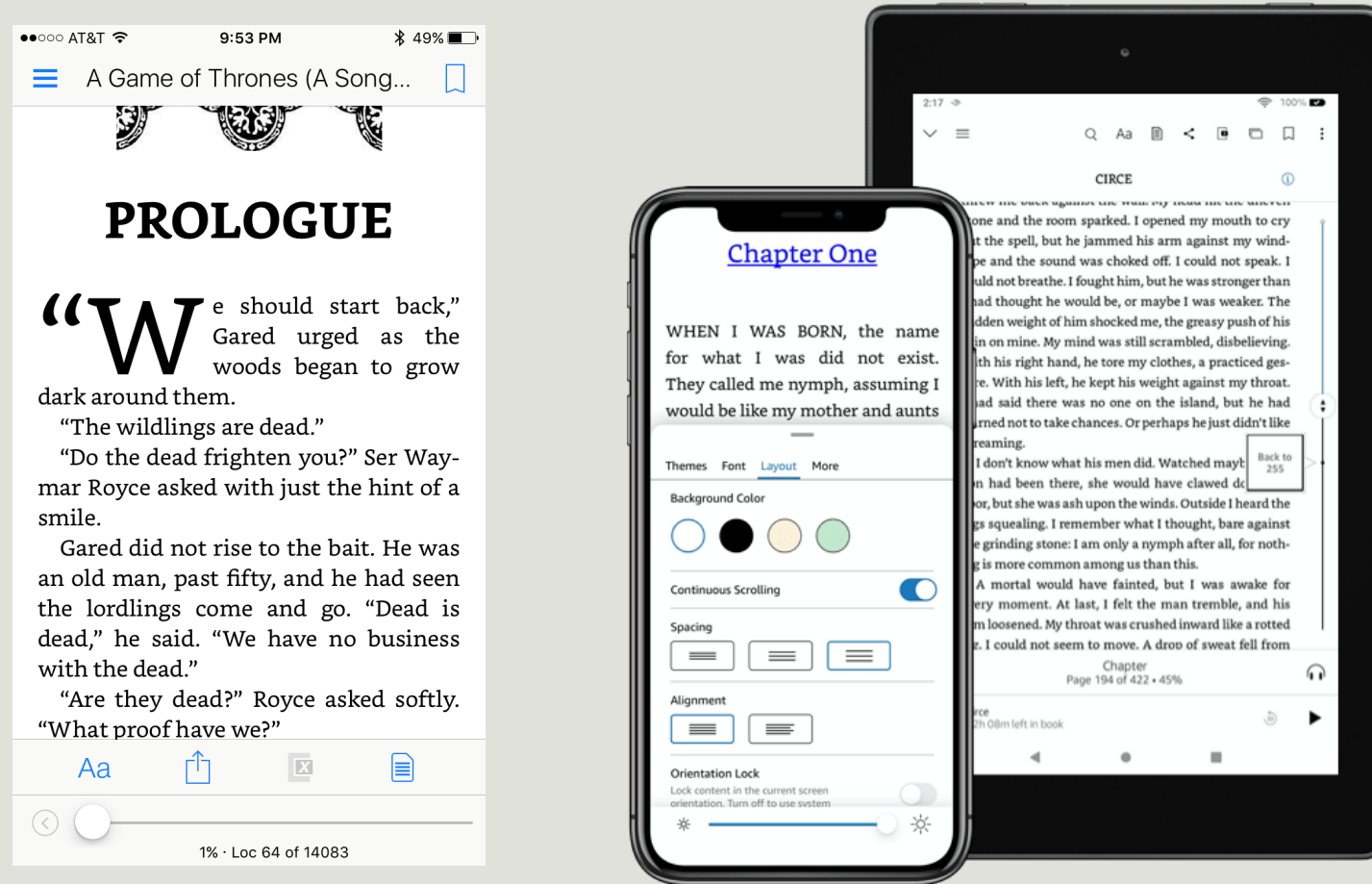
Interaction Design Book (ID):

1. Effectiveness
2. Efficiency
3. Safety
4. Utility
5. Learnability
6. Memorability
7. Satisfaction

Learnability

- How easy a system is to learn to use (quick startup)
- Most important with high amount of novice users
- Key Questions:
 - Can the user figure out the system by exploring?
 - How hard will it be to learn the whole set of functionality?

High Learnability



Low Learnability

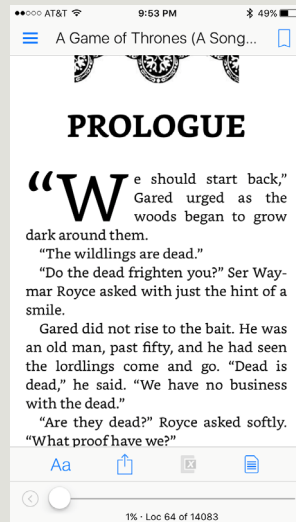


Measuring Learnability

How long does it take for a novice user to be able to do a series of set tasks?

Primary goals:

- Allow users to learn systems through exploration
- Make it easy for all users to learn the whole set of functionalities



Efficiency

- The way a product supports users in carrying out their tasks
- Once a user has learned the system, can they sustain a high level of productivity?
- Key Questions:
 - How many steps does it take to complete a task?
 - How does storing a person's personal details make it more efficient?

Low Efficiency

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N

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[Home Page](#) [Enrollment](#)

ENROLLMENT

STUDENT ACCOUNTS

FINANCIAL AID

ACADEMICS

ADMISSIONS

PROFILE

Fall 2023

BUILD SCHEDULE & ENROLL

TEXTBOOK ONLINE ORDERING

DROP

VIEW SCHEDULE

SHOPPING CART

COURSE	TITLE	DAY	TIME	LOCATION	CREDITS	INSTRUCTOR	STATUS
CSCE - 378 - 001	HUM-COMPUTR INTERACT	MWF	03:30PM - 04:20PM	LPH-102 CITY	3	Alisha Bevins	O

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SEARCH

Click [here](#) for classic enroll

VIDEOS

- **Beginner:** Scheduler Planner Basics
- **Intermediate:** Customizing Campus Options
- **Intermediate:** Using Locks to Narrow Down Schedules
- **Intermediate:** Customizing Parts of Term Options
- **Advanced:** Using Custom Course Options

INSTRUCTIONS

- Building your Schedule
- Search for Classes
- Class Details
- Add/Drop
- Changing Class Options

What is the average downtime per task? How many keystrokes/ buttons per task?

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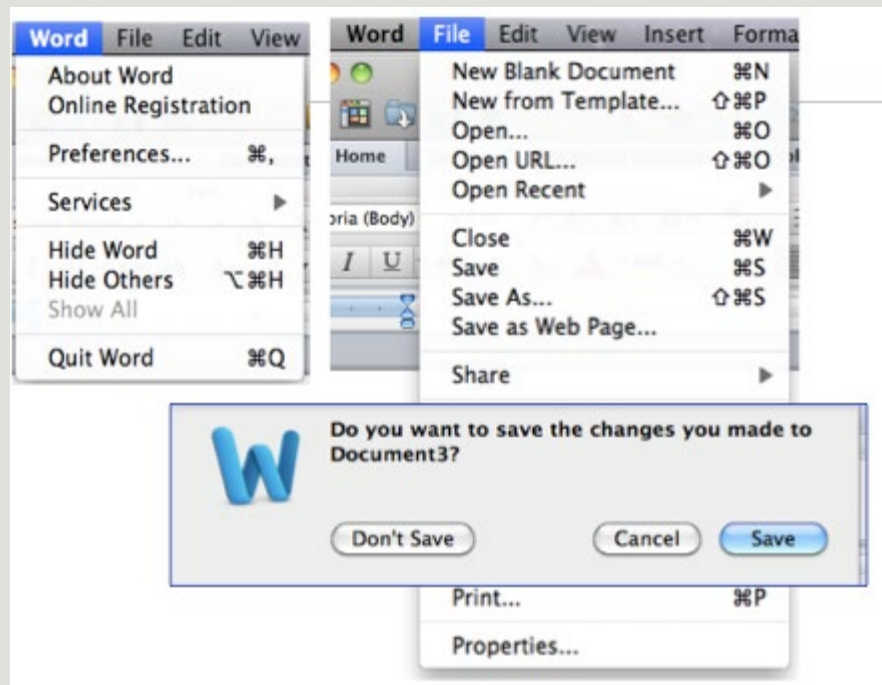
Errors (Safety)

Protecting the user from dangerous conditions and undesirable situations, also from perceived fear

Questions:

- What is the range of possible errors?
- What measures to recover easily from them?

Errors (High Safety)



Errors (Low Safety)



Measuring Errors (Safety)

How often does data get lost or do you choose the wrong option?

When this is most important:

- If a user consistently performs the wrong action
- When data gets lost often
- User fears negative consequences

